

by using XAML. The declarative approach is similar to the approach that is taken by ASP.NET in that it enables you to define event handlers in markup language. However, you must provide all of your event handler implementations imperatively by using managed code, in a file that is known as the code-behind file.

To handle WPF control events, you must first specify the event handler for the control in the XAML file and then implement the event handler in Visual Basic .NET, Visual C#, or any other language that is supported by the .NET Framework, in the code-behind file.

1. Specify the event handler in the XAML file

The following code example specifies an event handler for the Click event of a button in the XAML file.

```
<Button Name="Button1" Click="Button1_Click">
Click here
</Button>
```

2. Implement the event handler in the code-behind file

The following code examples implement an event handler for the Click event of a button in the code-behind file.

```
public void Button1_Click(object sender, RoutedEventArgs
e)
{
    MessageBox.Show("Hello WPF");
}
```

When you create an application that has a user interface, your application uses events to respond to the actions of the user. When the user interacts with an element in the user interface, for example, by clicking a button or selecting a value, that element fires an event. In your application code, you can create methods that are executed every time a particular event is fired. In this way, the actions of the user drive the behavior of your application.

In WPF applications, user interface elements are organized in a hierarchical structure that is known as the element tree. The Window or Page element is at the top of the element tree. The Window element can include one or more container controls, such as a StackPanel, and the StackPanel can include controls such as buttons. In this case, the button controls are at the bottom of the element tree. Routed events can propagate up or down the element tree. There are three types of routed events.